

Riccati-Padé quantization and oscillators $V(r) = gr^\alpha$

Francisco M. Fernández* and R. Guardiola

Departamento de Física Atómica y Nuclear, Universidad de Valencia, Avenida Dr. Moliner 50, 46100-Burjassot, Valencia, Spain

M. Znojil

Ústav jaderné fyziky Akademie věd České republiky, 250 68 Řež u Prahy, Czech Republic

(Received 10 June 1993)

We develop an alternative construction of bound states based on matching the Riccati threshold and asymptotic expansions via their two-point Padé interpolation. As a form of quantization it gives highly accurate eigenvalues and eigenfunctions.

PACS number(s): 03.65.Ge

I. POWER-LAW POTENTIALS

The nonrelativistic description of quarkonia [1] reattracted attention to the simple power-law potentials

$$V_\alpha(r) = Cr^\alpha. \quad (1)$$

This formula incorporates all the exactly and completely solvable forces listed by Newton [2] as special cases. We get the Mathieu-solvable dipolelike spike when $\alpha = -4$, a modified free motion when $\alpha = -2$, the Coulomb field when $\alpha = -1$, a harmonic oscillator when $\alpha = 2$, and a square well in the limit $\alpha \rightarrow \infty$.

Besides the quarkonium confinement models with a small positive $\alpha > 0$, phenomenological forces (1) with negative α 's may find applications as dipole or multipole (i.e., Lennard-Jones-like) $\alpha \neq -1$ modifications of the Coulombic interaction, or as simulated many-body screenings in mean-field theories of the Thomas-Fermi type [3].

Formally, the simplicity of forces (1) is of great appeal. Their semiclassical WKB treatment leads to a closed and compact form for the spectrum [4], and even perturbative considerations prove efficient in the broad vicinity of the above-listed set of exactly solvable exponents α [5–7].

The complicated extrapolation integer- $\alpha \rightarrow$ noninteger- α of Ref. [5] proves redundant for the “dense subset” of exponents

$$\alpha = \frac{I_1}{I_2} > -2, \quad I_1, I_2 = (\text{integers}). \quad (2)$$

Indeed, via the Liouville change of variables $r \rightarrow r^{\text{const}}$ [8], the underlying radial Schrödinger equation

$$\Psi''(r) = \left(V(r) + \frac{l(l+1)}{r^2} - E \right) \Psi(r) \quad (3)$$

can be transformed into an equivalent bound-state problem with integer α 's ($I_2' \rightarrow 1$, [9]). Thus the phenomenologically desirable choice of rational exponents (2) may *a priori* be made without any significant expected growth of complications. Similarly, we might even further extend our scope and consider superpositions of forces with exponents given by (2). In the latter case, if needed, a few particular *exact* bound states could be constructed in an elementary form [10] or generalized to the so-called Hill-determinant expansions with certain “minimally non-numerical,” quasi-special-function structure [9]. For simplicity and to facilitate the discussion here we concentrate on Eq. (1) and in particular on the case $\alpha = -1/2$.

II. THE RICCATI METHOD

Equation (3) is formally an ordinary linear differential equation of second order. For suitable potentials $V(r)$, this equation defines bound states and their spectrum of energies E_n , $n = 0, 1, \dots$ ($E_n < 0$ for $\alpha < 0$), once we satisfy the standard boundary conditions near the origin,

$$\Psi(r_0) = 0, \quad r_0 \ll 1 \quad (4)$$

and at infinity,

$$\Psi(r_\infty) = 0, \quad r_\infty \gg 1. \quad (5)$$

The *Ansatz*

$$\Psi(r) = \exp \left[- \int^r f(x) dx \right] \quad (6)$$

or, in the other words,

$$f(r) = - \frac{\Psi'(r)}{\Psi(r)} \quad (7)$$

*Permanent address: QUINOR, Facultad de Ciencias Exactas, Universidad Nacional de La Plata, Calle 47 entre 1 y 115, Casilla de Correo 962, 1900 La Plata, Argentina.

converts Eq. (3) into its Riccati equivalent

$$f'(r) = f^2(r) - \frac{l(l+1)}{r^2} - V(r) + E \quad (8)$$

which is nonlinear. The subsequent Liouvillean change of variables,

$$r \rightarrow t = r^\beta, \quad (9)$$

transforms Eq. (8) into an equivalent differential Riccati-type equation for the function $g(t) \equiv f(r(t))$,

$$tg'(t)/\beta = t^\beta g^2(t) - l(l+1)t^{-\beta} + Et^\beta - V(r(t))t^\beta, \quad (10)$$

which is better suited for our purposes.

Our choice of the example (1) supports the choice $\beta = I_2$ or $\beta = -I_2$ in Eq. (9). In the former case we have $r = t^{I_2}$ that transforms Eq. (10) into

$$\begin{aligned} \frac{1}{I_2} t^{I_2+1} g'(t) &= t^{2I_2} g^2(t) - l(l+1) \\ &+ Et^{2I_2} - C(\alpha) t^{(\alpha+2)I_2}, \end{aligned} \quad (11)$$

which contains just integer powers of the variable. Analogously, if $r = t^{-I_2}$ we obtain

$$-\frac{1}{I_2} t^{I_2+1} g'(t) = g^2(t) - l(l+1)t^{2I_2} + E - C(\alpha)t^{-\alpha I_2}. \quad (12)$$

$$\begin{aligned} g_0 &= -l-1, \quad g_1 = g_2 = g_5 = 0, \quad g_3 = 1/(2l+5/2), \quad g_4 = E/(2l+3), \\ g_6 &= 1/[(2l+5/2)^2(2l+4)], \quad g_7 = 2E/[(2l+5/2)(2l+3)(2l+9/2)], \\ g_8 &= E^2/[(2l+3)^2(2l+5)], \quad g_9 = 1/[(2l+5/2)^3(2l+4)(2l+11/2)], \\ &\vdots \\ &\vdots \end{aligned} \quad (15)$$

and

$$\begin{aligned} G_0 &= \sqrt{-E} = k = 1/(2\rho) > 0, \quad G_1 = -\rho, \quad G_2 = -\rho^3, \\ G_3 &= \frac{1}{2}\rho^2 - 2\rho^5, \quad G_4 = 2\rho^4 - 5\rho^7 + l(l+1)\rho, \dots \end{aligned} \quad (16)$$

Their form follows from the pair of recurrence relations

$$g_j = \frac{1}{j/2 + 2l + 1} \sum_{k=3}^{j-3} g_k g_{j-k}, \quad j = 6, 7, \dots \quad (17)$$

and

$$G_j = (1 - j/2)\rho G_{j-2} - \rho \sum_{k=1}^{j-1} G_k G_{j-k}, \quad j = 5, 6, \dots, \quad (18)$$

derived by mere insertion of Eqs.(13) and (14) into the respective Schrödinger-Riccati dynamical equations (11) and (12).

Now, both equations can be solved by the power-series method.

For the sake of definiteness, let us restrict our attention to $\alpha = -1/2$, i.e., $I_1 = -1$ and $I_2 = 2$. Without loss of generality we will consider the value $C = -1$ for the coupling constant. Denote $s = \sqrt{r}$, i.e., $r(s) = s^2$. Then, near the origin, Eq. (11) may be solved by the *Ansatz*

$$f(r(s)) = \sum_{j=0}^{\infty} g_j s^{j-2}. \quad (13)$$

By analogy, we also postulate

$$f(r(s)) = \sum_{j=0}^{\infty} G_j s^{-j} \quad (14)$$

for Eq. (12) near infinity.

We have not yet taken into account the boundary conditions in the construction of $f(r)$. For this reason we still have a pair of independent solutions. Indeed, the proper insertions give the ambiguity $g_0 = -l-1$ or $g_0 = l$ in Eq. (13) and $G_0 = k$ or $G_0 = -k$, $k = \sqrt{-E} > 0$, in Eq. (14). The boundary conditions (4) and (5) determine the physical coefficients. In the present case, the “physical” choices read $g_0 = -l-1$ and $G_0 = k > 0$, and expansions (13) and (14) coincide with the so-called regular and Jost solutions, respectively [2].

The first few nonzero coefficients remain simple,

III. PADÉ INTERPOLATION

The construction of bound states is virtually completed. We simply have to combine our regular-solution series (13) and (15) with the asymptotic boundary condition (5) or, alternatively, complement the Jost solution (14) and (16) with the appropriate boundary condition (4) near the origin. Numerical experience leads us to prefer a third possibility, namely, to match the logarithmic derivatives of the regular and Jost solutions at some suitably chosen intermediate point $r_1 = O(1)$. For our example, we display both leading-order terms in our bound-state wave functions in an explicit form,

$$\Psi(r) = r^{l+1} e^{-kr} \exp \left[- \int^r F(x) dx \right], \quad (19)$$

which reminds one of the exactly solvable examples or, in a broader methodical setting, of the so-called method of Hill determinants [9].

In the recurrently studied class of the partially solvable

potentials (see Ref. [10] for a concise review of the related literature), our “eikonal” or “action” exponent function $F(x)$ always degenerates to a ratio of two polynomials. Now, let us generalize or rather “invert” this observation and postulate that for *all* the bound states the integrands are well approximated by the analogous formula,

$$\Psi'(r(s))/\Psi(r(s)) \approx \frac{\sum_{j=0}^M a_j s^j}{\sum_{i=0}^N b_i s^i} \equiv [M/N](s),$$

$$s \in (0, \infty), \quad (20)$$

i.e., mathematically, by a two-point Padé approximant [11].

Formally, the requirement (20) of mutual compatibility of the regular and Jost solutions may be written as a pair of equations,

$$[M/N](s) = \begin{cases} \sum_{j=-2}^J g_{j+2} s^j + O(s^{J+1}), & s \ll 1 \\ \sum_{j=0}^L G_j s^{-j} + O(s^{-L-1}), & s \gg 1. \end{cases} \quad (21)$$

$$(22)$$

The most appealing feature of this alternative form of quantization is its easy technical realization. Indeed, Eq. (13) implies that $b_0 = b_1 = 0$ while Eq. (14) requires that $M = N$. Equations (21) and (22) force the remaining coefficients to satisfy

$$a_n = \sum_{j=2}^{\min(N, n+2)} b_j g_{n+2-j}, \quad n = 0, 1, \dots, J+2 \quad (23)$$

and

$$a_{N-n} = \sum_{j=0}^{\min(N-2, n)} b_{n-j} G_{n-j}, \quad n = 0, 1, \dots, L, \quad (24)$$

with $a_{N+1} = a_{N+2} = \dots = 0$ and $a_{-1} = a_{-2} = \dots = 0$.

In our basic Padé formula (20) one of the coefficients fixes an arbitrary normalization and the remaining depend on the energy E . The number of free parameters, $2N$, must be equal to the $J + L + 4$ conditions (23) and (24). As long as $N + 1$ conditions are needed to define the a_n 's in terms of the b_m 's we are left with only $N - 1$ linear algebraic equations for b_m 's. It is well known that nontrivial solutions of the latter homogeneous set of equations exist if and only if its secular determinant vanishes.

The explicit form of this secular equation depends on the mutual relationship between J and L . Thus we get

$$H_{L,J}(E) = \begin{vmatrix} (g_2 - G_0) & g_3 & \cdots & g_N \\ g_3 & g_4 & \cdots & g_{N+1} \\ \cdots & \cdots & \cdots & \cdots \\ g_N & g_{N+1} & \cdots & g_{2N-2} \end{vmatrix} = 0, \quad (25)$$

for $L = 0$. Similarly, we may write

$$H_{L,J}(E) = \begin{vmatrix} (g_1 - G_1) & (g_2 - G_0) & \cdots & g_{N-1} \\ (g_2 - G_0) & g_3 & \cdots & g_N \\ g_3 & g_4 & \cdots & g_{N+1} \\ \cdots & \cdots & \cdots & \cdots \\ g_{N-1} & g_N & \cdots & g_{2N-3} \end{vmatrix} = 0, \quad (26)$$

for $L = 1$, and so on. We end up in the opposite extreme

$$H_{L,J}(E) = \begin{vmatrix} -G_{2N-2} & -G_{2N-3} & \cdots & -G_N \\ -G_{2N-3} & -G_{2N-4} & \cdots & -G_{N-1} \\ \cdots & \cdots & \cdots & \cdots \\ -G_N & -G_{N-1} & \cdots & (g_0 - G_2) \end{vmatrix} = 0, \quad (27)$$

with the minimal nontrivial choice of $J = -2$.

IV. NUMERICAL TESTS AND DISCUSSION

We may go one step further and consider $L = -1$ with one-point Padé formulas and secular equation

$$H_{L,J}(E) = \begin{vmatrix} g_3 & g_4 & \cdots & g_{N+1} \\ g_4 & g_5 & \cdots & g_{N+2} \\ \cdots & \cdots & \cdots & \cdots \\ g_{N+1} & g_{N+2} & \cdots & g_{2N-1} \end{vmatrix} = 0. \quad (28)$$

It is surprising that even without any *a priori* guarantee of validity of the asymptotic boundary conditions (5), Eq. (28) yields physical energies in the limit $N \rightarrow \infty$. As the so-called Riccati-Padé method, this amazing phenomenon was recently discovered, described and thoroughly tested in Refs. [12–16]. It certainly deserves a further study [17]. Here, we shall use it as a source of exact reference energies. Table I shows that in fact the boundary conditions are satisfied *a posteriori* up to a very good precision.

In an opposite extreme, to less surprise, the one-point Padé $J = -3$ condition

$$H_{L,J}(E) = \begin{vmatrix} -G_{2N-1} & -G_{2N-2} & \cdots & -G_{N+1} \\ -G_{2N-2} & -G_{2N-3} & \cdots & -G_N \\ \cdots & \cdots & \cdots & \cdots \\ -G_{N+1} & -G_N & \cdots & -G_3 \end{vmatrix} = 0 \quad (29)$$

TABLE I. Ground-state eigenvalues E and convergence of the one-point Padé extrapolated ratios a_N/b_N to their correct physical value $\sqrt{-E}$ in the missing-boundary-condition $L = -1$ case.

N	E	a_N/b_N	$(a_N/b_N)^2$
2	-0.4598	0.4567	0.209
3	-0.4362	0.7479	0.559
4	-0.43818	0.6362	0.4048
5	-0.438031	0.6704	0.4494
14	-0.4380412419425060	0.6618465	0.4380408
15	-0.43804124194250588	0.66184695	0.438041385

TABLE II. Rate of convergence of the one- and two-point Padé ground-state energies. The number of correct digits $p = -\log_{10} |E_{\text{approx}} - E_{\text{exact}}|$ is tabulated as a function of the numbers $L + 1$ and $J + 3$ of terms in the matched large- r and small- r expansions.

$\frac{1}{2}(J + L + 4)$	One-point Padé		Two-point Padé		One-point Padé
	$L = -1$	$L = 0$	$L = 1$	$L = 2$	$J = -3$
2		0.76	1.02	1.09	1.09
3	1.66	1.76	1.88	1.99	2.11
4	2.73	2.81	2.87	2.91	2.64
5	3.84	3.88	3.91	3.92	2.77
6	4.98	4.99	4.99	4.97	2.82
7	6.13	6.12	6.10	6.06	2.79
8	7.31	7.28	7.24	7.18	2.77
9	8.50	8.45	8.40	8.33	2.69
10	9.68	9.67	9.56	9.52	2.64

fails. The last column of Table II may be consulted as an illustration.

The main part of Table II samples the precision of our two-point Padé prescription. The number of correct digits is characterized by the logarithmic difference $p = -\log_{10} |E_{\text{approx}} - E_{\text{exact}}|$ in which the reference energy E_{exact} is taken from the last row of Table I. We see that for small values of L the input information about the small- r and large- r behavior proves to be of comparable importance. This phenomenon has its natural boundaries. With an excess of information from the large- r expansion we would lose the advantage of the small- r – large- r balance and return slowly to the oscillatory or even entirely divergent results as sampled by the very last column of Table II.

Similar conclusions are drawn about the wave functions $\Psi(r)$. Table III verifies their accuracy *directly* by the evaluation of the local energy $H\Psi/\Psi$ which is close to E provided that Ψ is close to the actual eigenfunction. In Table III, based on the [6/6] Padé approximants to $f(r)$, we compare the simple one-point formulas (first column) with their subsequent two-point Padé descendants. We would like to call the reader's attention to the flatness of $H\Psi/\Psi$ for the higher values of L which exhibits deviations smaller than one part in 10^7 or 10^8 . Again, the incorporation of information from the large- r region does not improve the results indefinitely, and an optimal L is expected in practical applications.

We conclude that the two-point Padé algorithm works in full accord with our *a priori* expectations. Intuitively, the details of the classically forbidden region $r \gg 1$ are

expected to be much less relevant for the shift of energies in comparison with the classically allowed domain of small r 's. Thus our results confirm observations about the WKB calculations [18].

Summarizing, our paper generalizes the Riccati-Padé method of Refs. [12–16] and improves its efficiency with regard to the wave functions. The use of small values of L proves sufficiently accurate for the calculation of energies. Independently, such a strategy may also be validated in the semiclassical WKB language, where energies also remain virtually independent of the details of forces and wave functions in the classically forbidden region.

In such a setting, our present method may also be viewed as a replacement of the WKB solutions, in both the classically allowed and forbidden regions, by the small- r and large- r expansions. We matched them via the straightforward two-point Padé prescription. In this way the present approach might complement the WKB method in the domain of the low-energy states. In accord with our numerical tests, it gives eigenvalues and eigenfunctions for these states with unprecedented accuracy. Numerical calculations for higher values of L suggest an oscillatory divergence, a thorough analysis of which lies beyond the scope of the present study.

Note added. Our choice of the class of forces (1) may prove unnecessarily restrictive in some important applications. For example, in accord with our first Ref. [1], even the deeper theory of heavy quarkonia seems to support a slightly extended class of models

$$V_{\alpha,\beta}(r) = Cr^\alpha + Dr^\beta. \quad (30)$$

TABLE III. Local energy $H\Psi(r)/\Psi(r)$ given by the two-point Padé [6/6] formula, as a function of the coordinate r .

L	-1	0	1	2	3	4
$\sqrt{r}$						
0	-0.438 042 0	-0.438 040 5	-0.438 042 0	-0.438 040 4	-0.438 042 2	-0.438 040 1
10	-0.437 660 8	-0.438 121 5	-0.438 025 5	-0.438 043 6	-0.438 041 6	-0.438 040 2
20	-0.436 905 9	-0.438 155 7	-0.438 030 8	-0.438 041 4	-0.438 042 1	-0.438 040 1
30	-0.436 381 4	-0.438 150 8	-0.438 035 0	-0.438 040 8	-0.438 042 2	-0.438 040 1
40	-0.436 028 4	-0.438 140 0	-0.438 037 3	-0.438 040 6	-0.438 042 2	-0.438 040 1
50	-0.435 779 3	-0.438 129 4	-0.438 038 7	-0.438 040 5	-0.438 042 2	-0.438 040 1

Its separate short- and long-range components may possess exponents, say, $\alpha = -1$ and $\beta = 1$ or, in general,

$$-2 < \alpha = \frac{I_1^{(1)}}{I_2} < 0 < \beta = \frac{I_1^{(2)}}{I_2} \quad (31)$$

with the other suitable integers $I_1^{(j)}$ and a common denominator I_2 .

Due to their phenomenological relevance, superpositions (30) as well as their more-term descendants have been widely studied in the recent literature. With $\alpha = -1$ and $\beta = 2$, Hautot [19] was probably the first to discover their partial exact solvability, nine years before Magyari [20] formulated the necessary and sufficient conditions of this remarkable phenomenon, and five years before Harrell [21] picked up similar oscillators in his non-standard perturbative considerations. In the present approach, the transition from our simple power-law Eq. (1) to the more flexible models (30) remains entirely straightforward: The same basic equation (10) leads to mere slight modifications of explicit relations (11) and (12) and *Ansätze* (13) and (14).

Due to their simplicity, the weakly spiked harmonic oscillators [i.e., forces (30) with the fixed $\beta = 2$ and variable negative exponents $\alpha > -2$] are of particular theoretical as well as phenomenological interest (cf., e.g., [22] for a short list of recent references). In the present context, their attractive case ($C = -1$) with $\alpha = -1/2$ would lead to mere single-item and fairly high-index modification

$$g_8^{(D \neq 0)} = [-D + E^2/(2l+3)^2]/(2l+5)$$

of our above $D = 0$ formula (15) while, needless to say, the same recurrences (17) will still apply (from $j = 9$ on, of course).

The repulsive alternative with $C = 1$ does not possess any confining $D = 0$ predecessor at all. Hence the related $D > 0$ modification of our above sequence of g_j 's will already encounter a change (namely, the change of sign) at g_3 . Moreover, due to the $r \gg 1$ asymptotic dominance of the new, $D \neq 0$ harmonic-oscillator term in the potential, our old $D = 0$ *Ansatz* (14) will change more thoroughly: Its sum must already start at $j = -2$.

Obviously, even the more pronounced changes in the latter solutions are still purely technical. Thus, omitting further details, we may summarize that our present formalism does not encounter any remarkable difficulties during its applications to the more-term power-law forces which are at most weakly singular in the origin.

ACKNOWLEDGMENTS

R.G. acknowledges the Dirección General de Investigación Científica y Tecnológica (DGICYT, Spain) for the Grant No. PB90-0417. F.M.F. acknowledges the Conselleria de Cultura, Educació i Ciència of the Generalitat Valenciana for financial support within the program Plan de Fomento a la Investigación Científica y Técnica, 1992. M. Z. acknowledges the Grantová agentura AV ČR, Praha for the Grant No. 14811.

-
- [1] C. Quigg and J. L. Rosner, *Phys. Rep.* **56**, 167 (1979).
 - [2] R. G. Newton, *Scattering Theory of Waves and Particles* (Springer-Verlag, New York, 1982).
 - [3] L. D. Landau and E. M. Lifshitz, *Quantum Mechanics* (GITTL, Moscow, 1963) (in Russian).
 - [4] F. Constantinescu and E. Magyari, *Problems in Quantum Mechanics* (Pergamon, New York, 1971), p. 106.
 - [5] C. M. Bender *et al.*, *Phys. Rev. D* **37**, 1472 (1987).
 - [6] V. C. Aguilara and R. Guardiola, *J. Math. Phys.* **32**, 2135 (1991).
 - [7] M. Znojil, *Phys. Lett. A* **164**, 145 (1992).
 - [8] F. W. J. Olver, *Introduction to Asymptotics and Special Functions* (Academic, New York, 1974), Chap. VI.
 - [9] M. Znojil, *J. Math. Phys.* **24**, 1136 (1983).
 - [10] M. Znojil and P. G. L. Leach, *J. Math. Phys.* **33**, 2785 (1992).
 - [11] G. A. Baker, Jr., *Essentials of Padé Approximants* (Academic, New York, 1975), p. 100.
 - [12] F. M. Fernández, G. I. Frydman, and E. A. Castro, *J. Phys. A* **22**, 641 (1989).
 - [13] F. M. Fernández, Q. Ma, and R. H. Tipping, *Phys. Rev. A* **39**, 1605 (1989).
 - [14] F. M. Fernández, Q. Ma, and R. H. Tipping, *Phys. Rev. A* **40**, 6149 (1989).
 - [15] F. M. Fernández, *Phys. Lett. A* **166**, 173 (1992).
 - [16] V. C. Aguilara-Navarro, F. M. Fernández, R. Guardiola, and J. Ros, *J. Phys. A* **25**, 6379 (1992).
 - [17] F. M. Fernández and R. Guardiola (unpublished).
 - [18] N. Fröman and P. O. Fröman, *JWKB Approximation* (North-Holland, Amsterdam, 1965).
 - [19] A. Hautot, *Phys. Lett. A* **38**, 305 (1972).
 - [20] E. Magyari, *Phys. Lett. A* **81**, 116 (1981).
 - [21] E. M. Harrell, *Ann. Phys. (N.Y.)* **105**, 379 (1977).
 - [22] M. Znojil, *Phys. Lett. A* **169**, 415 (1992), with further references.